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Emergency Logistics Distribution Center Selection Using Fuzzy AHP and Fuzzy VIKOR Methods

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Abstract: Disasters that directly affect human health negatively affect the supply chain. The construction of emergency centres is critical so that aid can quickly reach people in the region where disasters occur. By choosing the most appropriate location, transportation and logistics costs can be effectively reduced, emergency aid can be delivered faster, and the negative effects of disasters can be minimised. Thus, location selection research is popular in many different fields. Multi-Criteria Decision-Making (MCDM) methods are widely used in location selection problems. This study aims to determine the best distribution centre for emergency logistics (EL) using hybrid MCDM methods. The study used seven criteria and 11 alternative locations in Kahramanmaraş province of Turkey. In the research, the criteria used to determine the potential locations of distribution centres were weighted with the fuzzy Analytic Hierarchy Process (AHP) method. The alternative locations were ranked using weighted criteria using the fuzzy Vlsekriterijumska Optimizacija I Kompromisno Resenje (VIKOR) method. The research emphasises the importance of considering different criteria and using MCDM methods for the location selection of distribution centres. The research results help disaster managers and policymakers decide to place humanitarian logistics distribution centres strategically.

Keywords: distribution center; emergency logistics; fuzzy AHP; fuzzy VIKOR; MCDM

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1 Introduction

In general, we live with the risk of disaster. However, it seems that most people are unprepared. As it is known, the timing of every disaster cannot be predicted in advance. While some disasters occur without warning, others may escalate over time. Additionally, not every disaster occurs at the same place and time. While disasters such as snowstorms or avalanches may occur in some regions, hurricanes may impact others. Disaster preparation studies can be carried out depending on the types and locations of disasters, and effective and efficient disaster preparation can be carried out depending on the disaster type and region (Lu 2017). Disaster is a broad term that refers to a wide range of conditions, from acts of terrorism to natural disasters such as earthquakes. In the 2020 World Disaster Report published by the International Federation of Red Cross and Red Crescent Societies (IFRC), it is stated that disasters are evaluated according to a threshold of importance and events that cause the death of at least 10 people and affect more than 100 people are defined as disasters (IFRC 2020). Many definitions of disaster have been made in the literature. However, it is evident that scholars provide different definitions and interpretations depending on their discipline (Al-Madhari and Keller 1997; Shaluf 2007; IFRC 2020). The common features of all these definitions can be expressed as instantaneousness, unpredictability, uncontrollability, severe destruction, human loss and suffering exceeding the coping capacity of the affected community (Math et al. 2015). Natural disasters that cause devastating damage and socioeconomic problems have shown an increasing trend in recent years. In addition, overcoming disasters requires great effort and assistance.

Disasters can be natural disasters such as earthquakes, floods, and storms or human-made disasters such as fire, war, and terrorism. Hazards can be categorized as geophysical, biological, or anthropogenic. In addition to all these features, one of the most important problems is that disaster management is difficult for society (Auzzir et al 2018). Therefore, it is necessary to take responsibility to protect societies by developing effective strategies (Sun et al 2020). The Emergency Event Database (EM-DAT) recorded 387 natural hazards and disasters on a global scale, leading to a devastating loss of 30,704 lives and affecting 185 million people in 2012. The economic consequences were staggering, reaching an estimated US\$ 223.8 billion. Additionally, approximately 800,000 people were reported to be living in officially organised temporary accommodation by June 2023 (CRED 2023). The year 2023 witnessed earthquakes that caused devastating effects on a global scale. The five most powerful earthquakes since 1900 killed approximately one million people. Countries on active fault lines are especially vulnerable to earthquakes. Türkiye faces a significant earthquake risk in the Mediterranean-Alpine-Himalayan belt, with a sizable

earthquake striking the country every 5 years. On the 6th of February 2023, a significant occurrence transpired when two major earthquakes struck the southeastern region of Turkey. Regrettably, this catastrophic event led to the tragic loss of around 51,000 lives (Erden et al 2024). Currently, informal settlements provide shelter to the vast majority affected by the earthquake. The increasing occurrence of disasters and their worldwide repercussions places the demand for aid to be delivered efficiently and effectively. In this context, capacity building aims to enhance the effectiveness of meeting the needs of disaster-affected populations and strengthen local organizations for subsequent humanitarian interventions (OECD 2017).

Emergency logistics (EL) plays a crucial role in disaster relief and is linked to minimizing damage and improving survival rates in affected regions (Kundu et al 2022; Xu et al 2024). EL emphasizes optimizing selection targets and enhancing efficiency, distinguishing it from commercial logistics, which primarily prioritizes cost and profit. This distinction is particularly significant given the unpredictability of disasters, which introduces a high degree of uncertainty in EL operations (Wang et al. 2023). EL requires strict standards for gathering, coordinating, loading, and distributing resources using up-to-date information. Therefore, improving the efficiency of EL management is a crucial task for governments (Liu and Chen 2012; Nawazish 2022).

EL is a distinct aspect that deals with the immediate protection of resources, personnel, and capital, especially in the presence of significant natural disasters, sudden public health emergencies, events endangering public safety, military conflicts, and other emergencies. This field of logistics undertakes the critical responsibility of collecting, distributing, and transporting essential materials efficiently and rapidly (He and Wang 2023). EL management has gained significant prominence internationally over the last few decades due to the impact of numerous global disasters. Global disasters have unquestionably increased the provision and distribution of aid. Therefore, the importance of EL remains both during a disaster and its aftermath (Kundu et al 2022).

The EL system's capacity to withstand challenges has been significantly reduced (Gao et al 2021; Ge et al. 2020) due to various issues identified during this process, including limited response capacity, inconsistent dispatching, and inadequate transportation capabilities (Spieske and Birkel 2021). The main purpose of emergency response is to deliver basic aid such as water, food, medical supplies and shelter to areas affected by disasters. The aim is to alleviate human suffering and reduce the number of casualties. In this context, hours after the 7.7 magnitude earthquake struck the Kahramanmaraş province of Türkiye, another earthquake measuring 7.6 in magnitude occurred in the region, resulting in significant damage and loss of life. The area affected by the earthquake spanned 500 km in diameter, impacting ten provinces directly (Yılmaz et al. 2023). In such situations, EL selection

becomes a critical issue, as it facilitates the rapid delivery of aid to disaster-stricken areas. Effective EL selection can mitigate negative impacts while simultaneously reducing costs and enhancing efficiency (Feng et al. 2023). Consequently, research on EL selection holds substantial theoretical and practical value. This paper focuses on warehouse location selection to improve the efficiency of disaster response and EL operations following the 6 February 2023 Kahramanmaraş earthquake in Türkiye. Disaster events in recent years have increased the strategic importance of the warehouse location in the emergency area. For this reason, it becomes necessary to carefully select the distribution centre to minimise the problems experienced during disaster response processes and ensure easy equipment access when needed.

However, it is essential to validate the applicability of the model through comparative analysis. The literature reveals a limited number of studies employing Multi-Criteria Decision-Making (MCDM) approaches for EL selection. These existing studies have applied various MCDM methods. Therefore, the current study aims to identify the most suitable distribution center using fuzzy MCDM methods, specifically Fuzzy AHP and Fuzzy VIKOR. The impact of disasters on the sector is examined to develop a framework for understanding and managing these impacts. The article discusses the fuzzy MCDM method in detail. When applied in the field of disaster logistics, this method helps effectively evaluate data containing uncertainty and complexity effectively. It also demonstrates the advantages of the fuzzy MCDM method and its potential for a more comprehensive understanding and management of disaster logistics in the service sector.

The rest of the study is organized as follows. In the second section, literature on EL location selection is included. The third part explains the fuzzy *Vlsekriterijumska Optimizacija I Kompromisno Resenje* (VIKOR) and fuzzy Analytic Hierarchy Process (AHP) methods used in the study. In the fourth section, the results of the study are given. In the fifth chapter, a general evaluation of the study is provided.

2 Literature Review

MCDM methods are widely used in many areas such as health, location selection, education, energy, agriculture, food, tourism, supplier selection and disaster management (Zhang and Guo 2016; Manyaga et al 2020; Ortiz-Barrios et al. 2020; Yu et al 2021; Gul and Guneri 2021; Tao and Wang 2023; Brodny and Tutak 2023; Magableh 2023; Alamoodi et al. 2023; Kumar and Pant 2023; Lan et al. 2023; Sahoo et al 2024; Nayyeri et al 2024; Gai et al. 2024; Aktas Potur et al. 2025). There are studies using MCDM methods in EL centres and humanitarian logistics centres selection problems (Esmelian et al. 2015; AlThabahi et al. 2020; Geng et al 2021; Çetinkaya et al 2022; Boyacı and Şişman 2022; Feng et al. 2023; Erden et al 2024; Eelagh and Abbaspour

2024; Li et al 2025; Yildirim and Sisman 2025; Alemdar 2025). Campbell and Jones (2011) determined the optimum supply location using a cost model in their study on material supply in disaster preparation. Xu et al. (2016) used a two-level model in a random fuzzy environment to highlight that the 72-h post-earthquake location guidance problem is an important element of post-earthquake EL. Schempp et al. (2019) determined the optimum location of rescue centres using mathematical models. Yilmaz and Kabak (2020) determined Istanbul's main and local distribution centre selection criteria using fuzzy AHP and TOPSIS. They determined security, infrastructure, cost, and transportation for the main distribution criteria and warehouse, cost, and security criteria for local distribution criteria. Feng et al. (2023) used Entropy, VIKOR and TOPSIS methods to select Emergency Logistics Centers (ELC) in a city in China. They determined that the population density criterion has the greatest weight and that near the New City Plaza is the best alternative. The comparison analysis made with the TOPSIS method also confirmed the ranking. Kant et al (2023) aimed to develop disaster logistics central location selection with the Fuzzy AHP and BWM method for the needs of victims and teams in case of flood. Erden et al (2024) investigated the factors affecting the selection of humanitarian aid logistics distribution centres close to natural disasters in the Sakarya province of Türkiye. The study focused on determining criteria such as population density, proximity to major highways and airports, accessibility to public transport, natural disaster risk and infrastructure suitability. Yildirim and Sisman (2025) applied GIS-based MCDM to identify suitable emergency assembly areas. Li et al. (2025) integrated fuzzy-AHP and ontology for earthquake emergency shelter selection. Alemdar (2025) assessed seismic risk in Istanbul's transport network via GIS-MCDM. Aktas Potur et al. (2025) conducted a bibliometric study highlighting MCDM trends in disaster management. Chang et al. (2023) presented a strategic approach to address shipping challenges in multiple distribution centres by considering distribution and location selection costs with the PageRank and TOPSIS methods. Literature provides insight into areas requiring further research regarding EL distribution centre selection and fuzzy multi-criteria decision-making methods.

A review of the existing studies shows a consensus that MCDM approaches are effective for complex and uncertain decision-making problems such as EL location selection. Most researchers agree that accessibility, infrastructure, cost, and safety are key criteria (Yilmaz and Kabak 2020; Feng et al. 2023; Erden et al 2024). Furthermore, incorporating fuzzy logic into MCDM is widely accepted as a useful way to manage uncertainty and linguistic judgments in disaster environments (Xu et al. 2016; Kant et al 2023). However, scholars differ in their methodological preferences. Some prioritise mathematical optimisation and cost-based approaches (Campbell and Jones 2011; Schempp et al. 2019), while others emphasise expert-based, qualitative evaluations under fuzzy conditions (AlThabahi et al. 2020; Boyacı and

Şişman 2022). Likewise, some studies favour objective weighting techniques such as Entropy and TOPSIS (Feng et al. 2023), whereas others rely on subjective expert judgment methods like fuzzy AHP (Kant et al 2023; Yılmaz and Kabak 2020). These differences suggest that model selection depends largely on data quality, decision-maker expertise, and disaster context.

Despite growth of the study, several research gaps remain. Most prior studies employ a single MCDM method, which may reduce result robustness. Research focusing on Türkiye especially at the is limited, and few studies integrate hybrid fuzzy methods combining weighting and ranking (e.g., fuzzy AHP–VIKOR). Moreover, comparative validation between fuzzy-based and traditional models is rarely explored. Therefore, this study contributes by applying a hybrid fuzzy AHP–VIKOR approach in the Kahramanmaraş & scdil; province, aiming to improve the precision and practicality of EL location decisions.

Based on the literature review, it can be concluded that the selection of locations for emergency logistics should not be solely entrusted to field an expert. Considering this, the following research hypotheses/questions can be proposed: Can fuzzy MCDM methods be effectively applied for the emergency location selection problem when there is insufficient numerical data? Should the decision regarding the selection of emergency logistics distribution sites be left to managers or politicians, or should fuzzy MCDM methods be employed, incorporating the insights of field an expert? This study aims to address these research questions. In summary, the literature in approaches to EL location selection and identifies clear methodological gaps.

3 Methodology

The criteria and alternatives for the study were established based on interviews with an expert in Kahramanmaraş province and a comprehensive review of the literature (Roh et al. 2015; Boltürk et al. 2016; Budak et al. 2020; Geng et al 2020; Ak and Acar 2021; Al Mohamed et al. 2023; Tasdemir 2024; Erden et al 2024; Korucuk et al. 2024). The expert consulted in this study is a highly qualified individual who is well-acquainted with the Kahramanmaraş region, which has a high earthquake risk, and serves as a team leader among other specialists in the field of disaster logistics. Therefore, only this expert's opinion was used in the analysis. (K1), geographical location (K2), infrastructure (K3), transportation (K4), climate suitability (K5), security (K6), and sustainability (K7) have been identified as criteria for the study. The alternatives for the research are determined as the districts of Kahramanmaraş province, namely Onikişubat (A1), Dulkadiroğlu (A2), Elbistan (A3), Afşin (A4), Türkoğlu (A5), Pazarcık (A6), Göksun (A7), Andırın (A8), Çağlayancerit (A9), Nurhak (A10), and Ekinözü (A11). In the study, the fuzzy AHP method was used to weigh the criteria, and the obtained

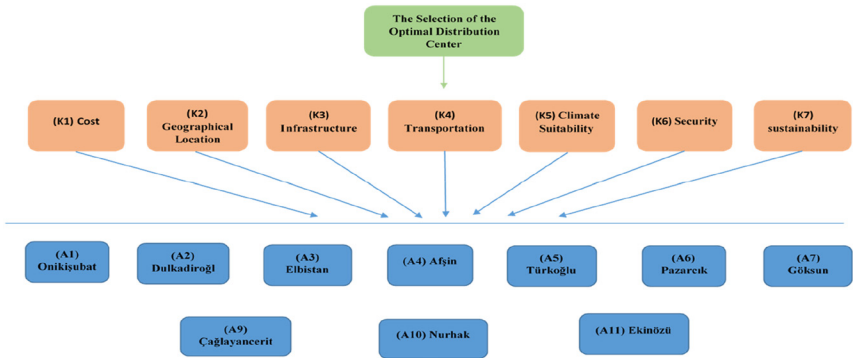


Figure 1: The Hierarchy of the research.

weights were included in the fuzzy VIKOR method to rank the alternatives. Figure 1 constructs the hierarchy of the research. At the top of the figure, the research objective is situated; in the middle section, the criteria employed in the research are delineated, and at the bottom, the alternatives considered in the study are presented. The fuzzy AHP and fuzzy VIKOR methods used in the study are explained below. Many fuzzy technologies have been introduced to MCDM models, significantly expanding the applications of MCDM methods. Among various fuzzy MCDM approaches, the fuzzy AHP and fuzzy VIKOR methods have a particularly broad range of applications (Kaya et al 2019; Liao et al. 2023). Fuzzy AHP and fuzzy VIKOR methods are among the most effective techniques for location selection (Tuğba Turğut et al. 2011; Ren and Lützen 2015; Boltürk et al. 2016; Otay 2023).

3.1 Fuzzy Analytic Hierarchy Process (AHP)

The Analytic Hierarchy Process (AHP) is a frequently utilised method for multi-criteria decision-making. It systematically establishes criteria weights and alternative priorities through structured pairwise comparisons. The AHP method is widely used worldwide. Over time, it is observed that some extensions of this method have been developed to address the shortcomings of AHP. Due to the potential uncertainty in subjective judgments during comparisons, combining fuzzy sets with AHP has led to Fuzzy AHP, which can deal with uncertainty and indecision (Liu et al 2020). Van Laarhoven and Pedrycz (1983) first introduced Fuzzy AHP, comparing fuzzy ratios defined by triangular membership functions. Buckley (1985) determined fuzzy priorities using trapezoidal membership functions by comparing fuzzy ratios using geometric mean. Chang (1996) utilised triangular fuzzy numbers.

$X = \{x_1, x_2, \dots, x_n\}$ forms the criterion set and $U = \{u_1, u_2, \dots, u_n\}$ forms an objective set. Degree analysis is applied for all criteria evaluated for the purpose, and with this analysis, m analysis values are obtained and all M_{gi}^j (triangular fuzzy numbers) values are represented as fuzzy (Kaptanoğlu and Özok 2010);

$$M_{gi}^1, M_{gi}^2, \dots, M_{gi}^m; i = 1, 2, \dots, n \text{ and } j = 1, 2, \dots, m$$

The steps for Chang's fuzzy AHP model are outlined below (Chang 1996; Ersoy and Doğan 2020; Ersoy 2021; Abdullah 2023; Arora and Gupta 2023).

Step 1: Fuzzy synthetic degree value for criterion i ;

$$S_i = \sum_{j=1}^m M_{gi}^j \otimes \left[\sum_{i=1}^n \sum_{j=1}^m M_{gi}^j \right]^{-1} \quad (1)$$

Step 2: $M_2 = (l_2, m_2, u_2) \geq M_1 = (l_1, m_1, u_1)$ the likelihood degree of the situation;

$$V(M_2 \geq M_1) = \sup_{y \geq x} [\min(\mu_{M_1}(x), \mu_{M_2}(y))] \quad (2)$$

Equality (2) expresses the likelihood degree of the event $M_2 \geq M_1$, and equation (3) illustrates the expansion of this expression.

$$V(M_2 \geq M_1) = \text{hgt}(M_1 \cap M_2) = \mu_{M_2}(d) = \begin{cases} 1, & m_2 \geq m_1 \\ 0, & l_1 \geq u_2 \\ \frac{l_1 - u_2}{(m_2 - u_2) - (m_1 - l_1)}, & \text{diğer} \end{cases} \quad (3)$$

To compare M_1 and M_2 , it is necessary to know the values of $V(M_2 \geq M_1)$ and $V(M_1 \geq M_2)$.

Step 3: The likelihood degree that a convex fuzzy number is greater than k fuzzy numbers M_i ($i = 1, 2, 3, \dots, k$) can be defined by equality (4).

$$\begin{aligned} V(M \geq M_1, M_2, \dots, M_k) &= V(M \geq M_1) \text{ ve } (M \geq M_2) \text{ ve } \dots \text{ ve } (M \geq M_k) \\ &= \min V(M \geq M_i), i = 1, 2, 3, \dots, k \end{aligned} \quad (4)$$

In this case, the following assumptions are made for S_j ,

$k = 1, 2, 3, \dots, n, k \neq i$ as follows;

$$d'(A_i) = \min V(S_i \geq S_k) \quad (5)$$

The weight vector is shown in equality (6). Here, A_i ($i = 1, 2, \dots, n$) consists of n elements.

$$W = (d'(A_1), d'(A_2), \dots, d'(A_n))^T \quad (6)$$

Step 4: The normalised weight vector W is shown inequality (7) through the normalisation process.

$$W = d(A_1), d(A_2), \dots, d(A_n)^T \quad (7)$$

Here, W is a non-fuzzy number.

3.2 Fuzzy VIKOR Method

The Fuzzy VIKOR method addresses fuzzy multi-criteria problems characterised by conflicting and incomparable criteria. It effectively tackles problems in a fuzzy environment, accommodating both fuzzy sets for criteria and weights. Fuzzy VIKOR empowers decision-makers to identify preferred solutions in real organizational scenarios with prevalent decision problems. The method relies on an aggregated fuzzy value that signifies the distance of an alternative to the ideal solution. In constructing the Fuzzy VIKOR algorithm, fuzzy operations and procedures are employed to rank fuzzy numbers (Opricovic and Tzeng 2007; Chen and Wang 2009). The Fuzzy VIKOR method consists of the following steps (Chen and Wang 2009; Ikram et al 2020; Kizielewicz and Bacziewicz 2021; Lam et al. 2021).

Step 1: Initially, decision-makers (n), alternatives (m), and criteria (k) are identified.

Step 2: includes the specification of linguistic variables and their corresponding triangular fuzzy numbers. These linguistic variables express the importance of criteria and rank alternatives based on various criteria.

Step 3: involves gathering opinions and preferences from decision-makers. The integrated fuzzy weight for each criterion is calculated using Equation (8), where n represents the number of decision-makers within this context.

$$\tilde{W}_j = \frac{1}{n} \left[\sum_{e=1}^n \tilde{W}_j^e \right], j = 1, 2, \dots, k \quad (8)$$

The importance weight of the i -th alternative concerning the j -th criterion is calculated using Equation (9).

$$\tilde{X}_{ij} = \frac{1}{n} \left[\sum_{e=1}^n \tilde{X}_{ij}^e \right], i = 1, 2, \dots, m \quad (9)$$

Step 4: The normalised fuzzy decision matrix is formed as follows.

$$C_1 \quad C_2 \cdots C_k$$

$$\tilde{D} = \begin{matrix} A_1 \\ A_2 \\ \vdots \\ A_m \end{matrix} \begin{bmatrix} \tilde{x}_{11} & \tilde{x}_{12} & \cdots & \tilde{x}_{1n} \\ \tilde{x}_{21} & \tilde{x}_{22} & \cdots & \tilde{x}_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \tilde{x}_{m1} & \tilde{x}_{m2} & \cdots & \tilde{x}_{mk} \end{bmatrix} \quad \tilde{W} = [\tilde{W}_1, \tilde{W}_2, \dots, \tilde{W}_k]$$

$$i = 1, 2, \dots, m; \quad j = 1, 2, \dots, k$$

$$\tilde{W} = [\tilde{W}_1, \tilde{W}_2, \dots, \tilde{W}_k] \quad j = 1, 2, \dots, k \quad (10)$$

Here $\tilde{x}_{ij}$, represents the degree of alternative A_i according to criterion C_j . $\tilde{W}_j$, expresses the importance weight of the criterion j .

Step 5: The fuzzy best value ($\tilde{f}_j^*$) and fuzzy worst value ($\tilde{f}_j^-$) are determined as shown in Equation (11).

$$\tilde{f}_j^* = \max_i \tilde{x}_{ij}, \tilde{f}_j^- = \min_i \tilde{x}_{ij} \quad (\text{Utility – based criteria}) \quad (11)$$

$$\tilde{f}_j^* = \min_i \tilde{x}_{ij}, \tilde{f}_j^- = \max_i \tilde{x}_{ij} \quad (\text{Utility – based criteria})$$

Step 6: The values of $\tilde{S}_i$ using equation (12) and $\tilde{R}_i$ using equation (13) are found.

$$\tilde{S}_i = \sum_{j=1}^k \tilde{W}_j (\tilde{f}_j^* - \tilde{x}_{ij}) / (\tilde{f}_j^* - \tilde{f}_j^-) \quad (12)$$

$$\tilde{R}_i = \max_j [\tilde{W}_j (\tilde{f}_j^* - \tilde{x}_{ij}) / (\tilde{f}_j^* - \tilde{f}_j^-)] \quad (13)$$

$\tilde{S}_i$, represents the sum of the distances of the i -th alternative (A_i) to the best fuzzy value for all criteria.

$\tilde{R}_i$, indicates the maximum distance of the alternative A_i to the fuzzy worst value for the j -th criterion.

Step 7: Calculate the $\tilde{S}^*$, $\tilde{S}^-$, $\tilde{R}^*$, $\tilde{R}^-$, $\tilde{Q}_i$ values as specified in Equations (14) and (15):

$$\tilde{S}^* = \min_i \tilde{S}_i, \quad \tilde{S}^- = \max_i \tilde{S}_i$$

$$\tilde{R}^* = \min_i \tilde{R}_i, \quad \tilde{R}^- = \max_i \tilde{R}_i \quad (14)$$

$$\tilde{Q}_i = v \frac{(\tilde{S}_i - \tilde{S}^*)}{\tilde{S}^- - \tilde{S}^*} + (1 - v) \frac{(\tilde{R}_i - \tilde{R}^*)}{\tilde{R}^- - \tilde{R}^*} \quad (15)$$

In this context, the symbol $\tilde{S}^*$, denotes the minimum value of the $\tilde{S}_i$ metric, signifying the maximum group benefit. Similarly, $\tilde{R}^*$ represents the minimum value of $\tilde{R}_i$,

indicating the least amount of opposing views. The $\tilde{Q}_i$ index is computed by assessing minimum regret and group benefit. Within this framework, the variable “ v ” denotes the weight assigned to the maximum group benefit strategy. Compromise can be achieved through strategies such as the “majority vote” ($v > 0.5$), as well as the “consensus” ($v = 0.5$) or “vote” ($v < 0.5$) strategies.

Step 8: The defuzzification of the triangular fuzzy number $\tilde{Q}_i$ is carried out, leading to the determination of the ranking of alternatives commencing from the smallest Q_i value. Various defuzzification strategies convert a fuzzy number into a crisp value. For this study, defuzzification was executed by calculating the averages of fuzzy numbers. The alternative with the smallest index value is considered the optimal choice.

Step 9: The two conditions that must be satisfied to obtain a compromise solution are presented in the following equalities.

Condition 1: Acceptable advantage refers to a condition designed to assess the presence of a substantial distinction between the optimal and the nearest alternatives. The rankings of alternatives a' and a'' in terms of the Q index are determined using Equations (16) and (17).

$$Q(a'') - Q(a') \geq DQ \quad (16)$$

$$DQ = \frac{1}{m-1} \text{ if } m \leq 4, DQ = 0,25 \quad (17)$$

Here, m represents the number of alternatives.

Condition 2: The minimum index value designates option ‘ a ’ as the preferred choice. This reconciliatory solution provides stability in the decision-making process.

If $Q(a^{(m)}) - (Qa') < DQ$ and the first condition cannot be satisfied, then $a^{(m)}$ and a' are considered similar reconciliatory solutions. Since the reconciliatory solutions ($a', a'', \dots, a^{(m)}$) are similar to each other, it is stated that a' does not possess a comparative superiority. If the second condition cannot be satisfied, a' has a comparative superiority, but decision-making lacks stability. Hence, the reconciliatory solutions of a' and a'' are indistinguishable.

Step 10: Finally, the best alternative is selected in this step. The minimum Q_i is chosen as the optimal solution.

4 Results

This study was carried out in Kahramanmaraş province of Türkiye. 11 districts of Kahramanmaraş, which were at the epicentre of the earthquake that resulted in the death of many people on February 6, 2023, were included in the study. The map of

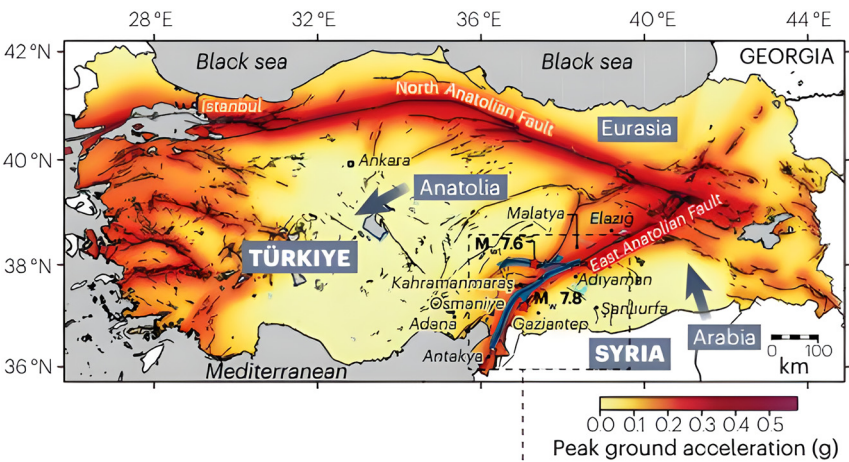


Figure 2: Türkiye earthquake map. Source: Hussain et al. 2023.

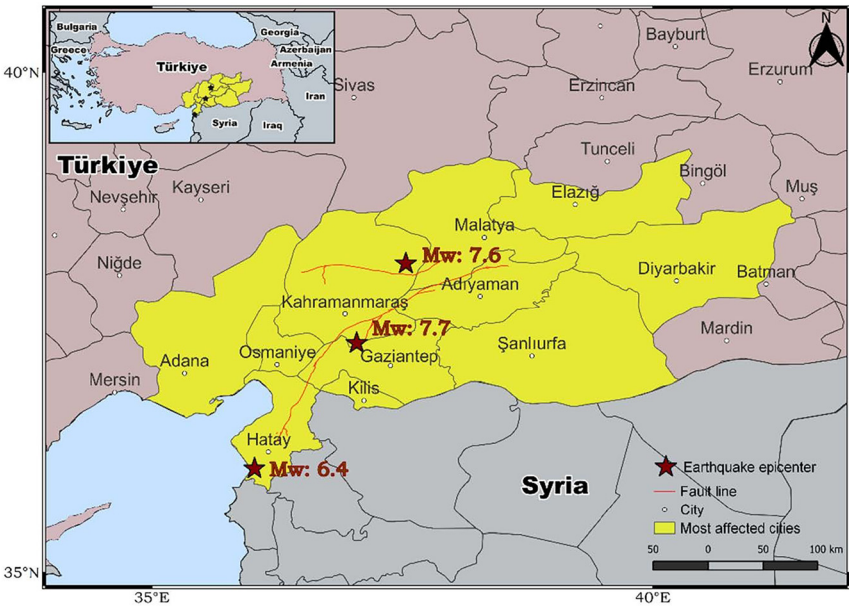


Figure 3: Areas affected by the Kahramanmaraş earthquake. Source: Ayso et al. 2025.

Türkiye is shown in Figure 2, areas affected by the Kahramanmaraş earthquake are shown in Figure 3, and the map of Kahramanmaraş is shown in Figure 4. The results of the study are given below.

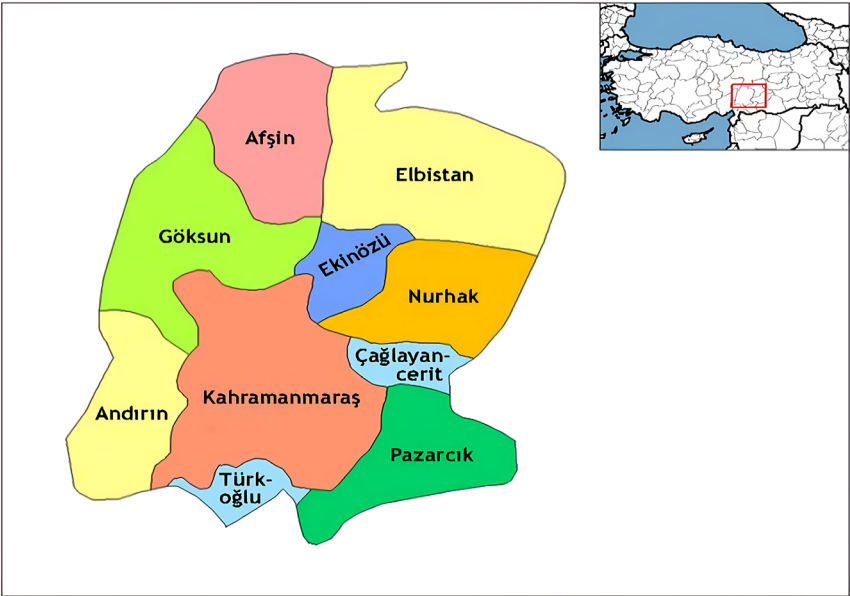


Figure 4: District of Kahramanmaraş. Source: tr.wikipedia.org 2024.

4.1 Results of Fuzzy AHP

In this study, the fuzzy AHP method was used to determine the weights of the criteria used to select distribution centres for EL. An expert decision-maker made pairwise comparisons of the criteria. The importance scale used in comparisons is presented in Table 1.

The matrices created to evaluate criteria were assessed using the fuzzy AHP importance scale in Table 1 with input from the expert decision-maker. After the

Table 1: Fuzzy AHP importance scale.

Explanation	Degree of importance	Degree of importance correspondence
Equally important	(1/2, 1, 3/2)	(2/3, 1, 2)
More important	(1, 3/2, 2)	(1/2, 2/3, 1)
Much more important	(3/2, 2, 5/2)	(2/5, 1/2, 2/3)
Very important	(2, 5/2, 3)	(1/3, 2/5, 1/2)
Definitely important	(5/2, 3, 7/2)	(2/7, 1/3, 2/5)

Table 2: Synthesis values of the criteria.

S_{K1}	0.07	0.07	0.09
S_{K2}	0.15	0.18	0.21
S_{K3}	0.06	0.07	0.08
S_{K4}	0.15	0.18	0.21
S_{K5}	0.12	0.14	0.17
S_{K6}	0.15	0.18	0.21
S_{K7}	0.15	0.18	0.21

Table 3: Importance weights for the criteria.

V_{K1}	V_{K2}	V_{K3}	V_{K4}	V_{K5}	V_{K6}	V_{K7}
0.00	1.00	1.00	1.00	1.00	1.00	1.00
1.00	1.00	0.00	1.00	0.81	1.00	1.00
0.00	1.00	0.00	1.00	1.00	1.00	1.00
0.00	1.00	0.00	1.00	0.81	1.00	1.00
0.00	1.00	0.00	1.00	0.81	1.00	1.00
0.00	1.00	0.00	1.00	0.81	1.00	1.00

evaluation, the steps related to Chang (1996) fuzzy AHP model were applied. Firstly, the synthesis values of the criteria are obtained according to Equation (1). The obtained synthesis values are presented in Table 2.

The synthesis values found in Table 2 are utilised to calculate the importance weights of the criteria, as per Equation (2) and (3). The results are presented in Table 3.

The priority values of the criteria were obtained using the importance weights (V) found in Table 3 with the application of Equation (5). The obtained priority values are as follows: $W = (0.00, 1.00, 0.00, 1.00, 0.81, 1.00, 1.00)$. The criterion weights resulting from the normalisation of priority values are presented in Table 4.

The criteria ranking based on the weights presented in Table 4 is as follows: $K2 = K4 = K6 = K7 > K5 > K1 = K3$. It is concluded that geographical location, transportation, security, and sustainability criteria share the same weight, indicating that they are considered the most important factors. Roh et al. (2015) employed the AHP method to weight five criteria in their study. The security criterion was ranked 3, with a weight of 0.20. In our study, the security criterion ranks either 1 or 3, with a weight of 0.21. Boltürk et al. (2016) utilized the Fuzzy AHP method to weight 5 criteria, assigning a weight of 0.21 to geographical location and 0.17 to transportation. Ak and

Table 4: Weights of the criteria.

Criterion number	Criter	Criterion weight
K1	Cost	0.00
K2	Geographical location	0.21
K3	Infrastructure	0.00
K4	Transportation	0.21
K5	Climate suitability	0.17
K6	Security	0.21
K7	Sustainability	0.21

Acar (2021) also applied the AHP method to weight 5 criteria, where the geographic location criterion was followed by the transportation and cost criteria in the ranking. Al Mohamed et al. (2023) determined the weights of 7 criteria using the Fuzzy AHP method, with the transportation criterion ranking first at a weight of 0.301. In our study, the transportation criterion ranks first, with a weight of 0.21.

4.2 Results of Fuzzy VIKOR

In this research, the criterion weights obtained through the fuzzy AHP method were incorporated into the fuzzy VIKOR method to rank the research alternatives. The importance scale used in the comparisons is presented in Table 5.

The comparison matrix of alternatives is presented in Table 6.

The fuzzy decision matrix presented in Table 6 has been transformed into crisp values using Equality (17) (Awasthi and Kannan 2016).

$$M = \frac{l + 4m + u}{6} \tag{18}$$

The resulting new matrix is presented in Table 7.

Table 5: Importance scale used in ranking alternatives.

Explanation	Degree of importance
Very weak	1, 1, 3
Weak	1, 3, 5
Moderate	3, 5, 7
Good	5, 7, 9
Very good	7, 9, 9

Table 6: Decision matrix for alternatives.

	K1			K2			K3			K4			K5			K6			K7		
A1	7	9	9	7	9	9	7	9	9	7	9	9	7	9	9	7	9	9	3	5	7
A2	7	9	9	7	9	9	7	9	9	7	9	9	7	9	9	7	9	9	3	5	7
A3	5	7	9	5	7	9	5	7	9	5	7	9	1	3	5	5	7	9	5	7	9
A4	5	7	9	5	7	9	5	7	9	5	7	9	1	3	5	5	7	9	5	7	9
A5	7	9	9	7	9	9	7	9	9	7	9	9	7	9	9	7	9	9	7	9	9
A6	7	9	9	7	9	9	7	9	9	7	9	9	7	9	9	7	9	9	7	9	9
A7	7	9	9	7	9	9	7	9	9	7	9	9	1	3	5	7	9	9	7	9	9
A8	1	3	5	1	3	5	1	3	5	1	3	5	1	3	5	5	7	9	5	7	9
A9	1	3	5	1	3	5	1	3	5	1	3	5	1	3	5	5	7	9	5	7	9
A10	1	3	5	1	3	5	1	3	5	1	3	5	1	3	5	5	7	9	5	7	9
A11	1	3	5	1	3	5	1	3	5	1	3	5	1	3	5	5	7	9	5	7	9

Table 7: Non-fuzzy decision matrix.

	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11
K1	8.67	8.67	7.00	7.00	8.67	8.67	8.67	3.00	3.00	3.00	3.00
K2	8.67	8.67	7.00	7.00	8.67	8.67	8.67	3.00	3.00	3.00	3.00
K3	8.67	8.67	7.00	7.00	8.67	8.67	8.67	3.00	3.00	3.00	3.00
K4	8.67	8.67	7.00	7.00	8.67	8.67	8.67	3.00	3.00	3.00	3.00
K5	8.67	8.67	3.00	3.00	8.67	8.67	3.00	3.00	3.00	3.00	3.00
K6	8.67	8.67	7.00	7.00	8.67	8.67	8.67	7.00	7.00	7.00	7.00
K7	5.00	5.00	7.00	7.00	8.67	8.67	8.67	7.00	7.00	7.00	7.00

In the non-fuzzy decision matrix presented in Table 7, the best f_j^* and the worst f_j^- values, as shown in Table 8, were calculated for each criterion using Equality (11).

The fuzzy AHP criterion weights presented in Table 4 have been incorporated into the fuzzy VIKOR method. The values of S_i and R_i , calculated according to Equations (12) and (13), are presented in Table 9.

Table 8: Best f_j^* and worst f_j^- values.

	f_j^*	f_j^-
K1	3.00	8.67
K2	8.67	3.00
K3	8.67	3.00
K4	8.67	3.00
K5	8.67	3.00
K6	8.67	7.00
K7	8.67	5.00

Table 9: S_i and R_i values.

	S_i	R_i
A1	0.21	0.21
A2	0.21	0.21
A3	0.59	0.21
A4	0.59	0.21
A5	0.00	0.00
A6	0.00	0.00
A7	0.17	0.17
A8	0.88	0.21
A9	0.88	0.21
A10	0.88	0.21
A11	0.88	0.21

Table 10: The values of S_i^+ , S_i^- , R_i^+ and R_i^- .

S_i^+	0.00
S_i^-	0.88
R_i^+	0.00
R_i^-	0.21

The values of S_i^+ , S_i^- , R_i^+ and R_i^- , calculated using Equality (14) are shown in Table 10.

The Q_i value has been calculated using Equality (15). In studies, the v value in Equation (15) is generally accepted as 0.5, and in this study, $v = 0.5$ has also been adopted. The Q_i value is presented in Table 11.

The ranking of the alternatives according to the Q_i (minimum) values in Table 11 is as follows: $A5 = A6 > A7 > A1 = A2 > A3 = A4 > A8 = A9 = A10 = A11$. According to this ranking, Türkoğlu (A5) and Pazarcık (A6) districts are identified as equally important alternatives. It was concluded that these two regions are the most suitable distribution locations in EL. When tested with equations (16) and (17), the difference between the top-ranked second alternative, Göksun (A7), and the first alternatives, Türkoğlu (A5) and Pazarcık (A6), was found to be 0.50. This value is greater than 0.25. Thus, condition one is satisfied. At this moment, it can be said that A5 and A6 have a comparative advantage. According to the Q index ranking, A5 and A6 are the best alternatives in terms of S and R index rankings. It has been concluded that Condition two is satisfied. As in previous studies, the most effective alternatives were identified (Roh et al. 2015; Boltürk et al. 2016; Geng et al 2020; Budak et al. 2020; Tasdemir 2024). The findings from the Fuzzy VIKOR method align with the results of these other studies.

To ensure the robustness of the ranking results, a basic sensitivity analysis was performed by slightly varying the criterion weights. Specifically, a 5 % decrease in the weight of Criterion 2, a 10 % decrease in Criterion 4, a 5 % increase in Criterion 6, and a

Table 11: Fuzzy VIKOR Results.

Alternatives	Q_i
A1	0.61
A2	0.61
A3	0.83
A4	0.83
A5	0.00
A6	0.00
A7	0.50
A8	1.00
A9	1.00
A10	1.00
A11	1.00

5 % increase in Criterion seven were applied. Under these variations, the top three alternatives (A5, A6, and A7) maintained their positions, indicating that the ranking is stable. Similarly, when Criterion two was increased by 5 %, Criterion four and Criterion five were each increased by 5 %, while Criterion six and Criterion seven were decreased by 10 %, no change was observed in the rankings of the top three alternatives. These results demonstrate that small variations in the criterion weights do not significantly affect the ranking outcomes, confirming the robustness of the proposed model.

5 Conclusions

5.1 Theoretical Implication

The current study proposes an MCDM approach to select the most suitable distribution center of EL and to address the location problem of distribution centers. The objective is to identify the optimal distribution center for EL in the Kahramanmaraş province of Türkiye by employing a hybrid method that combines fuzzy AHP and fuzzy VIKOR techniques. First, seven criteria affecting EL distribution center were identified, and the weights of these criteria were determined using the fuzzy AHP method. Then, rankings of 11 alternative locations for the distribution center were obtained using the VIKOR method. The criteria of sustainability, geographical location, cost, transportation, infrastructure, climate suitability, and security were weighted using fuzzy AHP. The key criteria for determining the location of the distribution center included geographical location, transportation, security, and sustainability. According to the results of the fuzzy VIKOR method, A5 (Türkoğlu) and A6 (Pazarcık) emerged as the best alternatives. Feng et al. (2023) used CRITIC and Entropy-based TOPSIS and VIKOR methods for the site selection of 13 different EL

alternatives. Their study indicated that the population density criterion holds the highest weight, while proximity to the city square ranks highest among the alternatives. There are limited studies in the literature addressing EL using the MCDM approach. However, this study's outcomes generally reflect trends identified in the existing literature (Esmaelian et al. 2015; Yılmaz and Kabak 2020; Boyacı and Şişman 2022; Feng et al. 2023; Fei et al. 2024). These results support and complement previous findings by showing that the identified key criteria and the application of a hybrid fuzzy MCDM approach provide a systematic and effective way to evaluate and rank alternative EL distribution center locations. Consequently, this study contributes to the literature by demonstrating the successful application of fuzzy MCDM methods to EL location selection problems.

Like many other studies, this study has some limitations. The most important is that it focuses on only one province. Another limitation is that the study was conducted in Türkiye. In addition, the research focuses solely on the earthquake-affected area. For future research, alternative MCDM methods could be applied instead of Fuzzy AHP and Fuzzy VIKOR. Different regions in Türkiye and other parts of the world could also be investigated. Future studies may incorporate additional criteria beyond those used in this research. Overall, this study underlines the importance of basing decisions regarding emergency location selection on methodologically rigorous research and comprehensive analysis. The findings provide valuable information and serve as a useful resource for researchers and practitioners.

5.2 Practical Implication

Natural disasters and catastrophic events can create significant logistical challenges and various issues for residents of affected regions. In such situations, it is crucial that aid reaches these areas swiftly and efficiently. The selection of an appropriate location for distribution centers is vital for ensuring rapid access to affected zones and the prompt delivery of assistance. In our contemporary era, the detrimental impacts of natural disasters, epidemics, and conflicts underscore the importance of emergency preparedness and the necessary precautions that must be taken. The major earthquake, especially centred in Kahramanmaraş and affecting 11 provinces of Türkiye, reveals that preventive measures against such disasters are not always implemented on time in some cases. Therefore, the location of the distribution centre is of great importance in ensuring the fastest and most effective response in any emergency. The intervention in a short time to effectively manage emergencies represents the process of promptly reaching the incident site and minimising the overall adverse effects of the disaster, particularly in terms of loss of life. This paper aims to select the most suitable distribution centre for EL in Kahramanmaraş province using a hybrid fuzzy AHP and fuzzy VIKOR methods. Cost, geographical location, infrastructure, transportation,

climate suitability, security, and sustainability criteria are weighted using fuzzy AHP. The important criteria in determining the location of the distribution centre were geographical location, transportation, security and sustainability. According to the fuzzy VIKOR method results, A5 (Türkoğlu) and A6 (Pazarcık) were identified as the best alternatives. The current study will help both governments and non-governmental organizations operating in the relevant field in terms of EL distribution center site selection. Knowing the most important criteria and alternative selection in EL site selection will allow for effective planning and decision-making. Natural disasters and events are instantaneous. However, they incur heavy costs that cannot be compensated afterwards. Therefore, taking precautions in advance and being prepared will allow for minimizing the damage. The findings of the current study are particularly valuable in this respect and provide guidance for practitioners.

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Data availability: Not included but available on request.

Appendix

This study was conducted in the province of Kahramanmaraş to determine the optimal distribution centre for disaster and emergency logistics. Providing accurate and sincere answers to the survey questions is crucial for the usefulness of the research results. The data collected in the survey will be used for scientific purposes and will not be shared with third parties. Thank you in advance for dedicating your valuable time to contribute to the research. We appreciate your efforts and wish you success in your endeavours.

Matrix 1: The comparison of variables located in rows with variables located in columns should be conducted pairwise using the importance scale for matrices. Binary comparisons should be made based on the columns. The importance scale provided below is utilised for this matrix.

Scale of Importance

Explanation	Degree of importance	Importance matching
Equally important	(1/2, 1, 3/2)	(2/3, 1, 2)
More important	(1, 3/2, 2)	(1/2, 2/3, 1)
Much more important	(3/2, 2, 5/2)	(2/5, 1/2, 2/3)
Extremely important	(2, 5/2, 3)	(1/3, 2/5, 1/2)
Definitely important	(5/2, 3, 7/2)	(2/7, 1/3, 2/5)

Cost (C1)	Geographical location (C2)	Infrastructure (C3)	Transportation (C4)	Climate adaptability (C5)	Security (C6)	Sustainability (C7)
Cost (C1)						
Geographical location (C2)						
Infrastructure (C3)						
Transportation (C4)						
Climate adaptability (C5)						
Security (C6)						
Sustainability (C7)						

Matris 2: For Matrix 2, the importance scale provided below is used. The evaluation of alternatives is based on the criteria in Matrix 2.

Scale of Importance

Explanation	Degree of importance
Very weak	1, 1, 3
Weak	1, 3, 5
Moderate	3, 5, 7
Good	5, 7, 9
Very good	7, 9, 9

C1	C2	C3	C4	C5	C6	C7
Onikişubat						
Dulkadiroğlu						
Elbistan						
Afşin						
Türkoğlu						
Pazarcık						
Göksun						
Andırın						
Çağlayancerit						
Nurhak						
Ekinözü						

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